APEX ANALYTICA
MANIFOLD

CAUSAL-GRAPH FRAGILITY ANALYTICS

Manifold for Insurance & Reinsurance

A causal-inference research terminal for catastrophe modeling, treaty design, and cascade analysis across the reinsurance tower.

CONFIDENTIAL
PREPARED FOR **AON**
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THE GAP

Where the current toolset stops.

- **CAT models fit historical event sets.** When the climate regime changes, the fit breaks. The model can describe the old regime, not the new one.
- **Cross-peril correlation is inferred from thin data.** The calibration fails exactly under stress, when coupling matters most.
- **Sanctions and treaty cascade live in scenario libraries.** They become additive bumps to an exposure number that doesn't know the underlying structure exists.
- **Stress tests give magnitude, not propagation.** When a primary fails, every treaty downstream doesn't fail at the same severity. The structure decides which links carry the cascade, and the structure isn't in the model.

Stress tests show how big.
Causal graphs show how it propagates.

Most risk frameworks have the first and skip the second. Manifold builds the second on top of the structure your existing models assume away. The team comes from inside financial modelling, not adjacent to it: ex-HSBC AI, reinsurance, JHU systems engineering.

THE PLATFORM

Four engines, one configurable platform.

SPIRTES · CAUSAL DISCOVERY

Infers causal structure from claim, exposure, and balance-sheet panels. PCMCI on lagged returns and exposure deltas surfaces dependencies the historical fit missed.

TARSKI · FORMAL VERIFICATION

Verifies treaty terms, regulatory constraints, and ILS / parametric trigger structures against the inferred graph. Red-lines connections that violate the stated conditions.

PEARL · COUNTERFACTUAL INTERVENTION

Do-calculus on the graph. Answers the question stress tests gesture at and don't resolve: if you cut this exposure, what does the cascade look like.

PARETO · CASCADE SIMULATION

Per-node criticality estimators (BOCPD, persistent homology, transfer entropy) and full cascade simulation across the tower under stress.

THE SCORING FRAMEWORK

ΩF. Five pillars, per-node fragility.

Every node in the graph (counterparty, peril, treaty, jurisdiction, line of business) gets a score on five orthogonal pillars. The composite ΩF runs with bootstrap confidence intervals, so the readout shows uncertainty alongside the point estimate.

I
IRREPLACEABILITY

Substitutability of a counterparty, facility, or treaty under a freeze. Common-collateral choke points.

R
RESTORATION LATENCY

Time to replace capacity after catastrophic failure. Recapitalisation speed under stress.

J
JURISDICTIONAL HAZARD

Sanctions cascade, regulatory regime shifts, currency-zone fragmentation, cross-border CCP fragility.

C
CASCADE LOAD

Downstream exposure depth through the reinsurance tower. Common-collateral fire-sale loops.

T
TAIL DEPTH

Distributional depth beyond VaR (99.9% / EVT regime). Correlation-collapse asymmetry.

WHERE IT LANDS AT AON

Five concrete use cases across reinsurance and capital advisory.

- **CAT model augmentation.** Surface the structural correlations the AIR, RMS, or KCC fit missed: common-peril infrastructure dependencies, climate-driven shifts in coupling, sub-portfolio cascade depth.
- **Treaty cascade analysis.** If a primary insurer or a key reinsurance partner fails, what happens across the tower. Pearl do-calculus produces the quantified cascade with per-node fragility readouts.
- **Climate non-stationarity stress testing.** Explicit causal model of climate-loss linkages with regime-shift detection on the time series. Produces a defensible disclosure narrative for TCFD, NAIC, or Solvency II.
- **ILS and parametric trigger structuring.** Tarski formally verifies trigger conditions against the graph. Quantifies basis risk with structural causality, not statistical correlation.
- **Counterparty cascade across reinsurance partners.** ΩF per partner ranks fragility under stress. Interdiction analysis quantifies the marginal benefit of hardening any single link.

WHAT A PILOT DELIVERS

Twelve weeks, one question, four phases.

A pilot scopes Manifold to one question your team is facing: a treaty cascade, a peril correlation review, a counterparty network, or an ILS trigger portfolio. The engagement runs in four phases.

PHASE 1 · WEEKS 1-2

Data intake and graph reconstruction. Your team hands over the data shape. We map it to the Manifold schema and return a first-pass causal graph.

PHASE 2 · WEEKS 3-6

Per-node fragility readouts. Every node scored across the five **ΩF** pillars with bootstrap CIs. Top-3 fragility nodes ranked.

PHASE 3 · WEEKS 7-9

Interdiction analysis. Pearl do-calculus quantifies the marginal effect of each candidate intervention: cut this exposure, harden this counterparty, restructure this treaty.

PHASE 4 · WEEKS 10-12

Written deliverable and workshop. 40–60 page report, executive summary, and an on-site or remote walkthrough. Configured Manifold instance left running for AON read-access.

THE TEAM

Built inside the discipline, not adjacent to it.

CEO & PRINCIPAL SCIENTIST

Junaid Ghauri

D.Eng. Johns Hopkins. Bayesian modelling, COVID-19 analytics, reinsurance risk. Ex-CTO MARK LABS. Authored the **Ω**-Robustness foundations.

HEAD OF RESEARCH

Georgios Korpas

Ph.D. Mathematics, Trinity College Dublin (visiting Stanford). Ex-HSBC AI and optimization. Quantum algorithms for collateral and continuous optimization.

AI RESEARCH SCIENTIST

Jeremy Kulcsar

Ex-HSBC AI Research. Causality, portfolio optimization, deep learning. Author of the **Ω**F five-pillar scoring framework. Ecole Centrale Paris.

SOFTWARE & RESEARCH

Daniel Mastropietro

20+ years across energy, telecom, finance, government, manufacturing. Reinforcement learning, optimization, interpretability. Doctoral candidate INP Toulouse. Fulbright alumnus.

HEAD OF OPERATIONS

Brynna Shale

NYU Stern (Business and Data Science). Ex-Goldman Sachs Capital Reporting. Operational strategy, customer success, contracting and finance.

ADVISORS

Three advisors anchoring the work in published risk science.

RISK MODELING ADVISOR

Dr. Gonzalo L. Pita

Associate Research Scientist, Johns Hopkins. Natural disaster risk, climate adaptation, infrastructure vulnerability. Former Senior Consultant at the World Bank.

SYSTEMS ENGINEERING ADVISOR

Dr. Takeru Igusa

Professor, Johns Hopkins. Systems science, applied math, epidemiology, resilience. NIH and CDC funded. Anchors the cascade and restoration pillars in the framework.

DIGITAL & AI ADVISOR

Dr. Arnesh Telukdarie

Professor, University of Johannesburg. AI, large-scale industrial-systems design, digital transformation in regulated industries.

WHERE THIS PLUGS IN AT AON

Four AON business lines and how Manifold lands in each.

REINSURANCE SOLUTIONS

Treaty optimisation and counterparty cascade.

PROBLEM Cross-treaty correlation and downstream cascade when a primary or partner fails. Capital relief miscalculated when couplings are statistical, not structural.

MANIFOLD Causal graph of the tower. **Ω**F per treaty and partner. Pearl interdiction quantifies the marginal benefit of hardening any single link.

IMPACT FORECASTING

Catastrophe modelling under climate non-stationarity.

PROBLEM Historical-fit CAT models break under regime change. Cross-peril correlation degrades exactly when stressed. Sub-portfolio cascade depth is opaque to the fit.

MANIFOLD Spirtes / PCMCI surface structural correlations the fit missed. BOCPD flags regime shifts. Layers onto Impact Forecasting, doesn't replace it.

AON SECURITIES

ILS structuring, parametric triggers, basis risk.

PROBLEM Trigger validation and basis-risk quantification rely on assumed correlations that fail under regime shifts. Investor disclosure needs structural narrative.

MANIFOLD Tarski formally verifies trigger conditions against the causal graph. Pearl quantifies basis-risk scenarios. Audit-trail for investor and rating-agency disclosure.

CLIMATE RISK ADVISORY

Disclosure, scenario modelling, capital advisory.

PROBLEM TCFD, NAIC, and Solvency II disclosures increasingly demand structural-causality narratives. Statistical climate-loss models don't carry that story.

MANIFOLD Causal model of climate-loss linkages with regime-shift detection. Written audit-trail. Answers "which channels" rather than just magnitude.

CONTINUING THE CONVERSATION

If any of this is useful, we'd welcome a longer conversation.

If a use case in this deck maps to something your team is actually working on, we'd be glad to set aside 30 minutes to dig in. We can configure Manifold ahead of the call against a domain you care about, so the conversation has something concrete to point at rather than slideware.

If the deck raised more questions than answers, that's also a good outcome. Send them over and we'll respond in writing before any meeting if that's easier.

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